

HACKATHON · BRUSSELS · 13 JUNE 2026

Challenge *Statements*

Energy Data Visualisation Hackathon
Commons Hub Brussels

AXES**3 Challenges****FORMAT****1-Day Hackathon****AUDIENCE****All profiles**

Challenges Overview

AXE 1 Electrification

How might we make the cost and source of your energy mix visible, and actionable, for a Brussels household?

AXE 2 Load Shifting & Myth Busting

How might we make Belgium's energy reality visible, and shift behaviour through data storytelling?

AXE 3 Open Challenge

How might data visualisation help households reduce their energy consumption?

AXE 1 — ELECTRIFICATION

How might we make the cost and source of your energy mix *visible, and actionable*, for a Brussels household?

A large share of Brussels citizens are hesitant to electrify (heat pumps, EVs, e-bikes, HEMS) not because the technology isn't ready, but because the financial case feels opaque, risky, or simply unbelievable. Many plan to stay with gas heating and combustion engine cars. The missing piece is not a subsidy: it's a clear, honest, visual comparison.

Go beyond the sticker price. Belgian electricity bills have three components: taxes & excises, grid fees, and the commodity price. While the first two are fixed, the commodity price can vary during the day, for flexible hourly price contracts. Evening peaks (17:00–21:00) are the grid's biggest challenge: cooking, heating, and EV charging all converge exactly when renewable production is at its lowest.

THINK CONCRETELY — COMPARE THE REAL TOTAL COST OF

- Gas boiler vs. heat pump over 10 years, at current and projected energy prices
- A petrol car (10L/100km) vs. a Poppy (car-sharing), an EV charged at noon, and an e-bike or Hopper charged at night — for a typical Brussels commute from Commons Hub (BXL Centre) to Rouge-Cloître
- Running a 40°C washing machine at 18:00 vs. 23:00, on a dynamic tariff with the 2027 Sibelga capacity tariff applied

Factor in grid fees, dynamic tariffs, maintenance, and realistic usage patterns. Show the crossover point. Show the uncertainty range. Make the ROI (= Return On Investment) case that a hesitant citizen can actually trust, and make the right moment to act obvious.

Your challenge: build a visual tool that turns energy cost data into clear, honest comparisons. Make electrification choices financially credible and daily consumption decisions actionable, for a non-technical Brussels household. With or without HEMS (= Home Energy Management System).

AXE 2 — LOAD SHIFTING & MYTH BUSTING

How might we make Belgium's energy reality *visible*, and shift behaviour through data storytelling?

Electrification is stalling, not because the technology isn't ready, but because misinformation is winning and the grid's daily reality remains invisible to most citizens. Two forces are at play: widespread myths block behavioural change, and a structural consumption-production mismatch that is often overlooked. Both can be tackled with the right visualisation.

THE MYTHS WORTH BUSTING, GROUNDED IN REAL DATA

- **"The grid can absorb any peak"** — False. Grid operators cannot store electricity at scale; real-time balance is mandatory. Demand-side tools and simple consumption planning matter more than emergency signals.
- **"1 MW of solar = 1 MW of nuclear"** — Misleading. Capacity factors differ drastically: ~92% for nuclear, ~23% for PV in Belgium. Deploying renewables faster and smarter is the answer — not dismissing them.
- **"I inject as much as I consume, so I pay nothing"** — False. Grid fees apply regardless. You inject at noon when prices are lowest, and consume in the evening when prices are highest.
- **"Alerts help people consume less when needed"** — False. Panic drives irrational behaviour. Clear dashboards and simple planning tools beat alarm systems every time.
- **"My rooftop solar panels will not lower my bill"** — False. Payback period in Brussels: 3.5–4.5 years. Over 25 years, a well-maintained installation delivers over 200% ROI. The financial case is solid.
- **"E-bikes consume a lot of electricity"** — False. 0.5–2 kWh per 100 km — 10 to 30 times less than an EV. For short urban trips with little to no load, the annual cost is €2–€4.

GO FURTHER — THE BIGGER PICTURE

How has Brussels' energy consumption evolved over the last 50 years? How does quality of life correlate with energy use? How has GDP tracked against oil consumption? These macro stories reframe the transition as inevitable — not ideological. Most of our energy use is still fossil fuels for heating, which is neither cost-efficient nor healthy.

THREE ANGLES TO TACKLE

- **The awareness gap.** Most households don't know when the grid is under stress. Design a visual that makes peak hours and grid imbalance instantly understandable — in a way that motivates voluntary action, not panic.
- **The appliance blindspot.** Most households don't know which appliances drive their consumption peaks. Visualise appliance-level load data to help a household identify and shift its top consumption spikes — turning abstract kWh into recognisable objects like a dishwasher or a heat pump.
- **The production-consumption mismatch.** Solar produces at noon; households consume in the evening. Visualise this daily gap in a way that makes load shifting compelling for individuals and local policymakers alike.

Your challenge: build a visual story, art from data, or a dashboard that makes Belgium's energy reality understandable and the act of shifting consumption feel simple, worthwhile, and grounded in fact. The target audience is a Brussels citizen hesitant about the energy transition. Make the data trustworthy, not preachy. Make it shareable.

AXE 3 — OPEN CHALLENGE

How might data visualisation *help households* reduce their energy consumption?

The two previous challenges tackle specific, well-defined problems. This one is deliberately open.

If you have a fresh angle on household energy consumption — a dataset you want to explore, a behaviour you want to shift, a story you want to tell — this is your space. The only constraint: your solution must be visual, grounded in real data, and meaningful to a non-technical household.

*Energy is complex. Bills are opaque. Choices feel overwhelming.
Good visualisation can cut through all of that. **Show us how.***